FACULTY OF PSYCHOLOGY
AND EDUCATIONAL SCIENCES

AUTOMATING RELATIONAL DERIVATION: TOOLS FOR VISUALIZATION AND TASK GENERATION

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HIGHLIGHTS

Researchers in various fields of psychological and computer science research agree that **relational abilities are a cornerstone of human cognition** and intelligence (McLoughlin et al., 2020), and **many procedures** have been developped **to measure it** (e.g., Colbert et al., 2020; Alexander et al., 2016; Raven & Raven, 2003, ...).

We developped a set of **open-source tools** to facilitate research on relational abilities in both humans and artificial systems, allowing users to:

1. **Automatically derive relations**
2. **Plot relational networks**
3. **Automatically generate syllogistic reasoning and match-to-sample tasks**

DERIVATION TABLES

Derivation tables store patterns of relational derivation (mutual and combinatorial entailment) for different combinations of relations.

→ Digital versions of the tables can be used for **automated relational derivation**

→ Could be used in machine learning research, or to **facilitate research** on relational abilities

Mutual Entailment

AxB	BxA
Same (S)	Same (S)
Different (D)	Different (D)
Opposite (O)	Opposite (O)
More than (M)	Less than (L)
Less than (L)	More than (M)
Is part of (P)	Contains (C)
Contains (C)	Is part of (P)
Before (B)	After (A)
After (A)	Before (B)

Combinatorial Entailment (Linear)

AxB \ BxC	Same (S)	Different (D)	Opposite (O)	More than (M)	Less than (L)	Is part of (P)	Contains (C)	Before (B)	After (A)
Same (S)	S	D	O	M	L	P	C	B	A
Different (D)	D	/	/	/	/	/	/	/	/
Opposite (O)	O	/	S	/	/	/	/	/	/
More than (M)	M	/	/	M	/	/	/	/	/
Less than (L)	L	/	/	/	L	/	/	/	/
Is part of (P)	P	/	/	/	/	P	/	/	/
Contains (C)	C	/	/	/	/	/	C	/	/
Before (B)	B	/	/	/	/	/	/	B	/
After (A)	A	/	/	/	/	/	/	/	A

Combinatorial Entailment (Many to one)

AxB \ CxB	Same (S)	Different (D)	Opposite (O)	More than (M)	Less than (L)	Is part of (P)	Contains (C)	Before (B)	After (A)
Same (S)	S	D	O	L	M	C	P	A	B
Different (D)	D	/	/	/	/	/	/	/	/
Opposite (O)	O	/	S	/	/	/	/	/	/
More than (M)	M	/	/	/	M	/	/	/	/
Less than (L)	L	/	/	L	/	/	/	/	/
Is part of (P)	P	/	/	/	/	/	P	/	/
Contains (C)	C	/	/	/	/	C	/	/	/
Before (B)	B	/	/	/	/	/	/	/	B
After (A)	A	/	/	/	/	/	/	A	/

Comparative Relations (Linear)

AxB \ BxC	Less than (<)	More than (>)	Part of (P)	Contains (C)	Overlaps (O)	Is overlapped by (Ob)	Less than or equal to (<=)	More than or equal to (>=)	Starts (S)	Is started by (Sb)	Finishes (F)	Is finished by (Fb)
Less than (<)	<	/	/	<	<	/	<	/	<	<	/	<
More than (>)	/	>	/	>	/	>	/	>	/	>	>	>
Part of (P)	<	>	P	/	/	/	<	>	P	/	P	/
Contains (C)	/	/	/	C	/	/	/	/	/	C	/	C
Overlaps (O)	<	/	/	/	/	/	<	/	O	/	/	/
Is overlapped by (Ob)	/	>	/	/	/	/	/	>	/	/	Ob	/
Less than or equal to (<=)	<	/	/	<	<	/	<	/	<=	<=	/	<
More than or equal to (>=)	/	>	/	>	/	>	/	>	/	>	>=	>=
Starts (S)	<	>	C	/	/	/	<	>=	S	/	P	/
Is started by (Sb)	/	>	/	C	/	Ob	/	>=	/	Sb	Ob	C
Finishes (F)	<	>	/	/	/	/	<=	>	P	/	F	/
Is finished by (Fb)	<	/	/	C	O	/	<=	/	O	C	/	Fb

Temporal Relations (Linear)

AxB \ BxC	Before (B)	After (A)	During (D)	Contains (C)	Overlaps (O)	Overlapped by (Ob)	Meets (M)	Met by (Mb)	Starts (S)	Started by (Sb)	Finishes (F)	Finished by (Fb)
Before (B)	B	/	/	B	B	/	B	/	B	B	/	B
After (A)	/	A	/	A	/	A	/	A	/	A	A	A
During (D)	B	A	D	/	/	/	B	A	D	/	D	/
Contains (C)	/	/	/	C	/	/	/	/	/	C	/	C
Overlaps (O)	B	/	/	/	/	/	B	/	O	/	/	/
Overlapped by (Ob)	/	A	/	/	/	/	/	A	/	/	Ob	/
Meets (M)	B	/	/	B	B	/	B	/	M	M	/	B
Met by (Mb)	/	A	/	A	/	A	/	A	/	Mb	Mb	
Starts (S)	B	A	C	/	/	/	B	Mb	S	/	D	/
Started by (Sb)	/	A	/	C	/	Ob	/	Mb	/	Sb	Ob	C
Finishes (F)	B	A	/	/	/	/	M	A	D	/	F	/
Finsished by (Fb)	B	/	/	4	O	/	M	/	O	C	/	Fb

RELATIONAL TASK GENERATION

SYLLOGISTIC PROBLEMS

Syllogistic reasoning is considered a hallmark of human intelligence (e.g., Aristotle) and is studied in behavior-analytic research (e.g., Cassidy et al., 2011; Colbert et al., 2020), cognitive neuroscience (e.g., Goel & Dolan, 2001) and computer science (e.g., Hummel & Holyoak, 2001; Eisape et al., 2024).

Automated relation derivation allows for generating syllogistic reasoning problems :

- Involving **various (combinations of) relations**
- With **user-specified** problem **variations** (number of premises, premise order; irrelevant premise, ...) and response format (open format, forced choice)

LEB is the same as BEX.
Is BEX the same as LEB?

GUB is before BIR.
BIR is before LOP.
Is GUB to BIR the same as BIR to LOP?

GUB is more than BIR.
BIR is more than LOP.
LOP is more than PAF.
Is PAF more than GUB?

LEB is the same as BEX.
BEX is opposite to PAF.
PAF is opposite to GOR.
GOR is the same as NIB.
GUK is the same as BEX
Is BUR the same as DAV?

MATCH-TO-SAMPLE PROCEDURES

Match-to-sample procedures are very prevalent in research on stimulus equivalence (e.g., Sidman, 1994), Perez et al., 2023) and relational frame theory (e.g., Steele & Hayes, 1991)

Automated relation derivation allows for generating MTS procedures

- **Presets like Steele & Hayes** (1992) or any **user-defined constellation of relations**
- **User-specified task parameters** (training order, comparison stimuli, number of trials, ...)

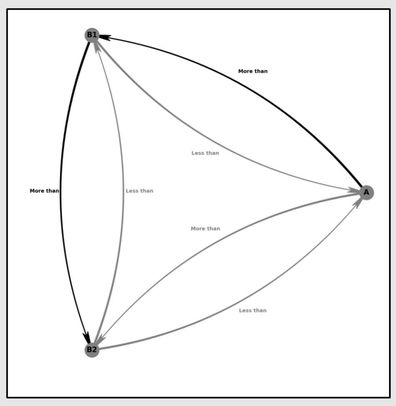
Training Baseline Relations												Mutual Entailment						Combinatorial Entailment											
(1)			(3)			(4)			(6)			(7)			(9)			(13)			(15)			(17)			(18)		
S			O			S			O			S			O			O			O			S			O		
A1			A1			A1			A1			B1			B3			B3			C3			B1			B1		
B1	B2	B3	B1	B2	*B3*	*C1*	C2	C3	*C3*	C1	C2	B2	*A1*	B3	*A1*	B2	N1	B2	*B1*	N1	N1	C2	*C1*	C2	*C1*	C3	C1	C2	*C3*

RELATIONAL NETWORK VISUALIZATION

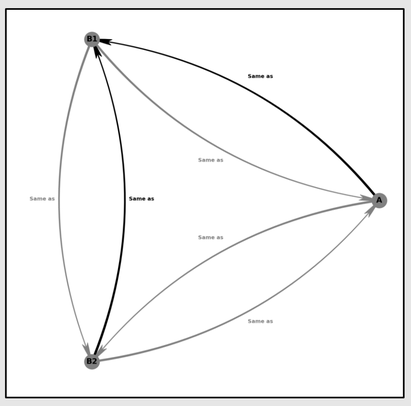
Automated relation derivation for plotting user-specified relational networks

- **Presets** like Steele & Hayes (1992) or **any user-defined relations**
- Plot baseline and derived relations separately or together and modify plot visuals

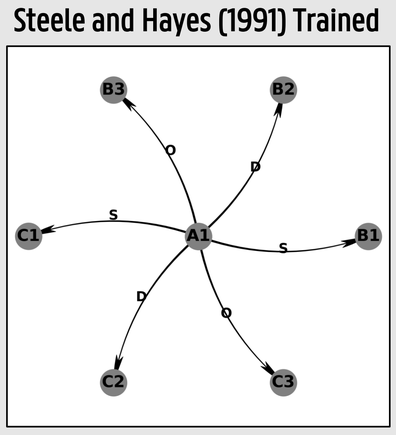
Linear More Than Relations



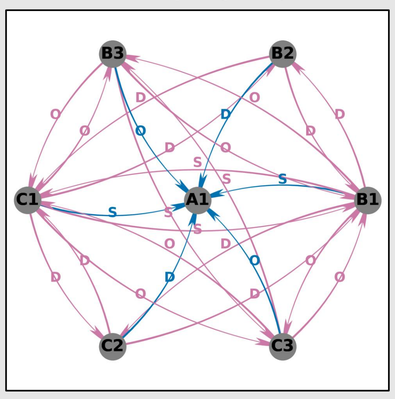
Many-to-One Sameness Relations



Steele and Hayes (1991) Trained



Steele and Hayes (1991) Derived



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